MDS - Microarchitectural Data Sampling - CVE-2018-12130, CVE-2018-12126, CVE-2018-12127, and CVE-2019-11091

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Resolved

Status

Important

Impact

Insights vulnerability analysis

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Executive Summary

Four new microprocessor flaws have been discovered, the most severe of which is rated by Red Hat Product Security as having an Important impact. These flaws, if exploited by an attacker with local shell access to a system, could allow data in the CPU's cache to be exposed to unauthorized processes. While difficult to execute, a skilled attacker could use these flaws to read memory from a virtual or containerized instance, or the underlying host system. Red Hat has mitigations prepared for affected systems and has detailed steps customers should take as they evaluate their exposure risk and formulate their response.

Issue Details and Background Information

Red Hat has been made aware of a series of microarchitectural (hardware) implementation issues that could allow an unprivileged local attacker to bypass conventional memory security restrictions in order to gain read access to privileged memory that would otherwise be inaccessible. These flaws could also be exploited by malicious code running within a container. These issues affect

many modern Intel microprocessors, requiring updates to the Linux kernel, virtualization stack, and updates to CPU microcode. The issues have been assigned CVE-2018-12130, as a severity impact of Important, CVE-2018-12126, CVE-2018-12127, and CVE-2019-11091 are considered Moderate severity.

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At this time, these specific flaws are only known to affect Intel-based processors, although Red Hat Product Security expects researchers to continue probing for unrelated Simultaneous Multi Threading (SMT) vulnerabilities across a wide range of vendors.

Flaws were found with the manner in which Intel microprocessor designs implement several performance micro-optimizations. Exploitation of the vulnerabilities provide attackers a side channel to access recently used data on the system belonging to other processes, containers, virtual machines, or to the kernel.

These vulnerabilities are referred to as Microarchitectural Data Sampling (MDS) due to the fact that they rely upon leveraging speculation to obtain state left within internal CPU structures.

CVE-2018-12126 - Microarchitectural Store Buffer Data Sampling (MSBDS)

A flaw was found in many Intel microprocessor designs related to a possible information leak of the processor store buffer structure which contains recent stores (writes) to memory.

Modern Intel microprocessors implement hardware-level micro-optimizations to improve the performance of writing data back to CPU caches. The write operation is split into STA (**ST**ore **A**ddress) and STD (**ST**ore **D**ata) sub-operations. These sub-operations allow the processor to hand-off address generation logic into these sub-operations for optimized writes. Both of these sub-operations write to a shared distributed processor structure called the 'processor store buffer'.

The processor store buffer is conceptually a table of address, value, and 'is valid' entries. As the sub-operations can execute independently of each other, they can each update the address, and/or value columns of the table independently. This means that at different points in time the address or value may be invalid.

The processor may speculatively forward entries from the store buffer. The split design used allows for such forwarding to speculatively use stale values, such as the wrong address, returning data from a previous unrelated store. Since this only occurs for

loads that will be reissued following the fault/assist resolution, the program is not architecturally impacted, but store buffer state can be leaked to malicious code carefully crafted to retrieve this data via side-channel analysis.

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The processor store buffer entries are equally divided between the number of active Hyper-Threads. Conditions such as power-state change can reallocate the processor store buffer entries in a half-updated state to another thread without ensuring that the entries have been cleared.

This issue is referred to by the researchers as Fallout.

CVE-2018-12127 - Microarchitectural Load Port Data Sampling (MLPDS)

Microprocessors use 'load ports' to perform load operations from memory or IO. During a load operation, the load port receives data from the memory or IO subsystem and then provides the data to the CPU registers and operations in the CPU's pipelines.

In some implementations, the writeback data bus within each load port can retain data values from older load operations until newer load operations overwrite that data

MLPDS can reveal stale load port data to malicious actors when:

- A faulting/assisting SSE/AVX/AVX-512 loads that are more than 64 bits in size
- A faulting/assisting load which spans a 64-byte boundary.

In the above cases, the load operation speculatively provides stale data values from the internal data structures to dependent operations. Speculatively forwarding this data does not end up modifying program execution, but this can be used as a widget to speculatively infer the contents of a victim process's data value through timing access to the load port.

CVE-2018-12130 - Microarchitectural Fill Buffer Data Sampling (MFBDS)

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This issue has the most risk associated, which Red Hat has rated as Important. A flaw was found by researchers in the implementation of fill buffers used by Intel microprocessors.

A fill buffer holds data that has missed in the processor L1 data cache, as a result of an attempt to use a value that is not present. When a Level 1 data cache miss occurs within an Intel core, the fill buffer design allows the processor to continue with other operations while the value to be accessed is loaded from higher levels of cache. The design also allows the result to be forwarded to the Execution Unit, acquiring the load directly without being written into the Level 1 data cache.

A load operation is not decoupled in the same way that a store is, but it does involve an Address Generation Unit (AGU) operation. If the AGU generates a fault (#PF, etc.) or an assist (A/D bits) then the classical Intel design would block the load and later reissue it. In contemporary designs, it instead allows subsequent speculation operations to temporarily see a forwarded data value from the fill buffer slot prior to the load actually taking place. Thus it is possible to read data that was recently accessed by another thread if the fill buffer entry is not overwritten.

This issue is referred to by researchers as RIDL or ZombieLoad.

CVE-2019-11091 - Microarchitectural Data Sampling Uncacheable Memory (MDSUM)

A flaw was found in the implementation of the "fill buffer," a mechanism used by modern CPUs when a cache-miss is made on L1 CPU cache. If an attacker can generate a load operation that would create a page fault, the execution will continue speculatively with incorrect data from the fill buffer, while the data is fetched from higher-level caches. This response time can be measured to infer data in the fill buffer.

Acknowledgements

Red Hat thanks Intel and industry partners for reporting this issue and collaborating on the mitigations for the same.

Additionally Red Hat thanks the original reporters:

Microarchitectural Store Buffer Data Sampling (MSBDS) - CVE-2018-12126

This vulnerability was found internally by Intel employees. Intel would like to thank Ke Sun, Henrique Kawakami, Kekai Hu and Rodrigo Branco. It was independently reported by Lei Shi - Qihoo - 360 CERT and by Marina Minkin, Daniel Moghimi, Moritz Lipp, Michael Schwarz, Jo Van Bulck, Daniel Genkin, Daniel Gruss, Berk Sunar, Frank Piessens, Yuval Yarom ([1]University of Michigan, Worcester Polytechnic Institute, Graz University of Technology, imec-DistriNet, KU Leuven, University of Adelaide).

Microarchitectural Load Port Data Sampling (MLPDS) - CVE-2018-12127

This vulnerability was found internally by Intel employees and Microsoft. Intel would like to thank Brandon Falk – Microsoft Windows Platform Security Team, Ke Sun, Henrique Kawakami, Kekai Hu, and Rodrigo Branco - Intel. It was independently reported by Matt Miller – Microsoft, and by Stephan van Schaik, Alyssa Milburn, Sebastian Österlund, Pietro Frigo, Kaveh Razavi, Herbert Bos, and Cristiano Giuffrida - VUSec group at VU Amsterdam.

Microarchitectural Fill Buffer Data Sampling (MFBDS) - CVE-2018-12130

This vulnerability was found internally by Intel employees. Intel would like to thank Ke Sun, Henrique Kawakami, Kekai Hu and Rodrigo Branco. It was independently reported by Giorgi Maisuradze – Microsoft Research, and by Dan Horea Lutas, and Andrei Lutas - Bitdefender, and by Volodymyr Pikhur, and by Stephan van Schaik, Alyssa Milburn, Sebastian Österlund, Pietro Frigo, Kaveh Razavi, Herbert Bos, and Cristiano Giuffrida - VUSec group at VU Amsterdam, and by Moritz Lipp, Michael Schwarz, and Daniel Gruss - Graz University of Technology.

Microarchitectural Data Sampling Uncacheable Memory (MDSUM) - CVE-2019-11091

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and Cristiano Giuffrida - VUSec group at VU Amsterdam.

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Additional References

For more information about this class of issues, please refer to Intel's Website

KCS: Simultaneous Multithreading in Red Hat Enterprise Linux

KCS: Disabling Hyper-Threading

KCS: CPU Side Channel Attack Index Page

KCS: Microcode availability for Pre-Haswell CPUs

KCS: Availability of Updated Intel CPU Microcode Addressing Microarchitectural Data Sampling (MDS) Vulnerability

KCS: Applying MDS CVE's patches on RHV hosts and manager node

KCS: RHOS Mitigation for MDS ("Microarchitectural Data Sampling") Security Flaws

Video: All about MDS in about 3 minutes

Video: Longform MDS Technical Explanation

Blog: Deeper Look at the MDS Vulnerability

Blog: Modern IT security: Sometimes caring is NOT sharing